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Design of a micro-UART for SoC application

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Abstract

This paper presents the design of a universal asynchronous receiver and transmitter (UART), which is fully functional and synthesizable. Due to its modularity, configurability and extremely compact size, the proposed UART is named as micro-UART and it is ideal for system-on-a-chip (SoC) application. The core is usable as an intellectual property. Verilog hardware description language (HDL) in the Altera's MAX-PLUS II environment has been used for its design, compilation and simulation. The UART has been implemented using Altera's FPGA technology.

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1. Introduction

There have been revolutions in semiconductor technology. Now multi-million transistors are being integrated on a single chip named as SoC to realize the full functionality of a system. SoC offers enormous advantages such as (i) improved performance (ii) reduction of cost (iii) increased reliability (iv) decreased power consumption (v) reduced system-size (vi) short-time to market etc.

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Usually predesigned intellectual property modules are stitched together in designing and developing SoC to reduce time-to-market. UART is a commonly used component in SoC for data communication. A micro-UART of compact size is now a burning demand in IC design industry.

UART from different companies such as NS16450A and NS16550A of National Semiconductor Company (NSC) [7], SC26C198 and SC28L92 from Philips Company are available in the market [2]. They are used as discrete component in designing electronic system. Design of UART chip has been discussed in some literature where it has been used as a platform to carry out experiments to solve different research problems [8,11,12]. Yeandel et al. [11] describe on-line testability issue while O'Neill et al. [8] highlight issues related to high speed serial communication and Yunshan and Marshall [12] implement IFIS (If it Fails It Stops) testability methodology using UART chip. The design of the UART chip presented in this paper claims uniqueness for its modularity feature and use of the industry standard Verilog HDL code. Verilog shortens the design cycle of a chip by efficiently describing its behavior [3,5,9]. It is technology independent. If a particular IC fabrication process becomes outdated, it is possible to synthesize a new level of design by only changing the synthesizing technology file but using the same Verilog HDL code.

The following sections of this paper describe the design, simulation and verification of the proposed micro-UART.

2. UART theory of operation

UART converts serial data to parallel and vice versa. A CPU communicates with a UART via its parallel interface while other peripheral devices communicate with a UART via its serial interface. It is named asynchronous receiver and transmitter because in the UART protocol, the transmitter and the receiver do not share the same clock signal. That is, a clock signal does not emanate from one UART transmitter to the other UART receiver. Fig. 1 shows block diagram of a UART. Its main components are transmitter module, receiver module, baud-rate generator and control circuit.

System-clock is used to produce baud_clock, which is generally 16 times baud-rate. Xmitter module converts parallel data input into serial data output and receiver module does vice versa.

In data transmission through UART, once the baud-rate has been established (prior to initial communication), both the transmitter and the receiver's internal clock are set to the same

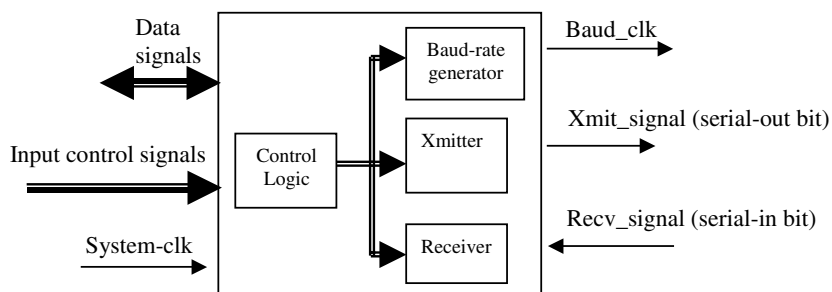


Fig. 1. Block diagram of a UART.

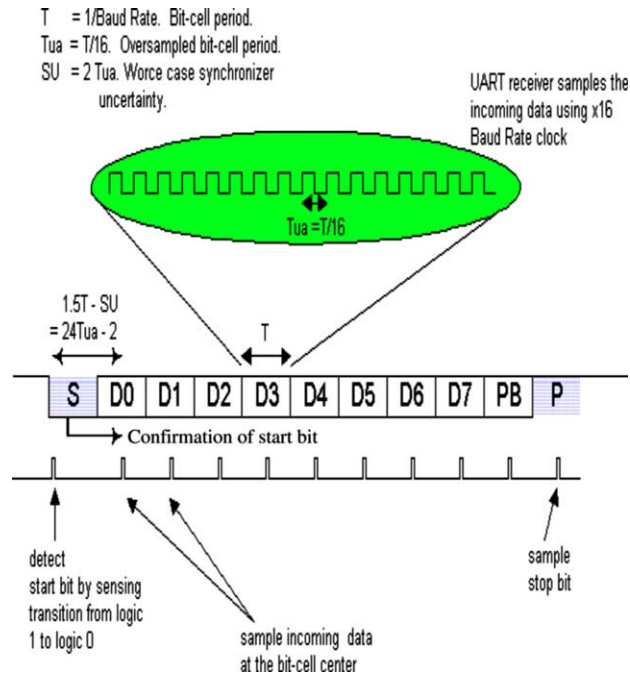


Fig. 2. Data sampling by UART.

frequency (though not the same phase). The receiver “synchronizes” its internal clock to that of the transmitter’s at the beginning of every data packet received. This allows the receiver to sample the data bit at the bit-cell center. A data packet is composed of 1 start-bit, which is always a logic 0, followed by a programmable number of data bits (typically between 6 and 8), an optional parity bit, and a programmable number of stop bits (typically 1 but it may be 1.5 or 2). The stop bit must always be logic 1. The receiver detects the start-bit by detecting the transition from logic 1 to logic 0 of the data line (the logic level of the data line is high while the UART is in idle state). Once the start-bit is detected, the next data bit’s “center” can be assured to be 24 ticks from the point of start-bit detection. From then on, every next data bit center is 16-clock ticks later. Fig. 2 illustrates this point.

3. Features of the proposed UART

The proposed UART is having the following features:

- Independently controlled transmitter and receiver module.
- Programmable baud-rate generator.
- Compact size.
- False start-bit detection.
- Programmable word length, 1 start-bit, no parity and 1 stop bit.

4. Design and simulation of the proposed UART

Design and simulation of the UART has been performed using Verilog under MAX-PLUS II environment [1,4,6,10]. All the functional modules of the UART have been designed and verified separately and then they have been integrated together.

Main modules of the UART are (i) baud-rate generator (ii) transmitter module (iii) receiver module.

4.1. Baud-rate generator

The UART contains a programmable baud-rate generator. It is capable of taking any clock input as long as PLD technology supports. The output frequency of the baud-rate generator is determined by the equation: **baud-clock = system-clock/baud * 16 * divisor** By changing the divisor value, frequency of the baud_clk can be changed to required value.

Fig. 3 shows the simulation result of the baud-rate generator for different divisor value. 'rst', 'bd_clk' and 'sys_clk' indicate the reset input signal, baud-clock signal and system-clock signal of the baud-rate generator.

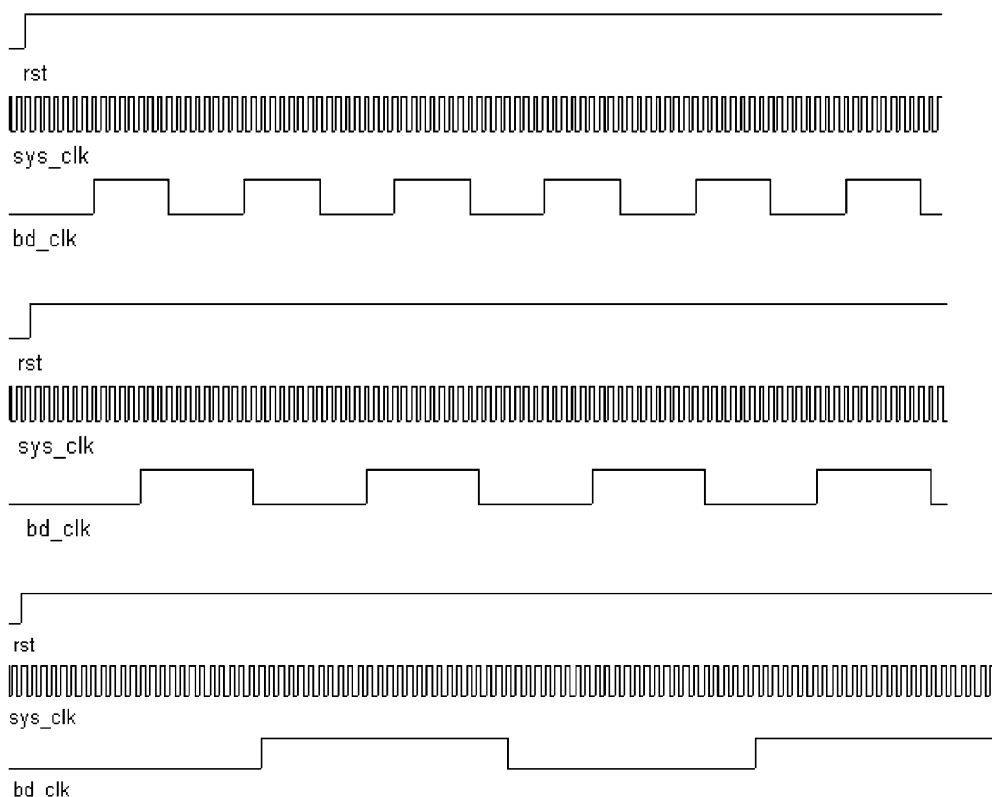


Fig. 3. Simulation result showing different bd_clk due to changing divisor value.

Fig. 3 shows that when reset input is high, baud-rate generator can produce baud clock of different frequency depending on the divisor value.

4.2. Transmitter module

Fig. 4 shows functional block diagram of the transmitter module. It consists of the following modules:

- finite state machine (FSM),
- shift-register,
- bit counter,
- event counter.

FSM controls all other modules. Shift-register converts 8-bit parallel input data to serial output data. Bit counter counts number of bits transmitted and event counter counts number of clock cycles. Clock input is equal to $16 \times \text{baud_rate}$ which is generated by baud_rate generator module.

Fig. 5 shows the state diagram of the transmitter module, which describes its operation. All the modules of the transmitter reset upon active low signal in the 'reset' input and the FSM defaults to **t_idle** state where the machine idles as long as 'start_pulse' input signal is low.

When there is an active high signal in the 'start_pulse' input, the shift-register is loaded with 8-bit data from the input 'data_in' and the FSM jumps to **t_start** state where 'tmit_data' output is set to '0' (start-bit) through a 2/1 mux. The FSM waits in this state for 16-clock cycle and then it jumps to **t_wait** state. In this state, 'tmit_dat' output is connected with serial output of the shift-register through the 2/1 mux. The FSM waits in this state for 16-clock cycle and then jumps to **t_shift** state where the shift-register is shifted by 1 bit and then jumps back to **t_wait** state. In this state, the FSM compares the number of transmitted bit. If number of transmitted bits are equal to presetted value (word_length) then the FSM jumps to **t_stop** state, otherwise it goes to **t_shift** state. In **t_stop** state, 'tmit_dat' output is set to '1' (stop bit) through 2/1 mux. The FSM waits in

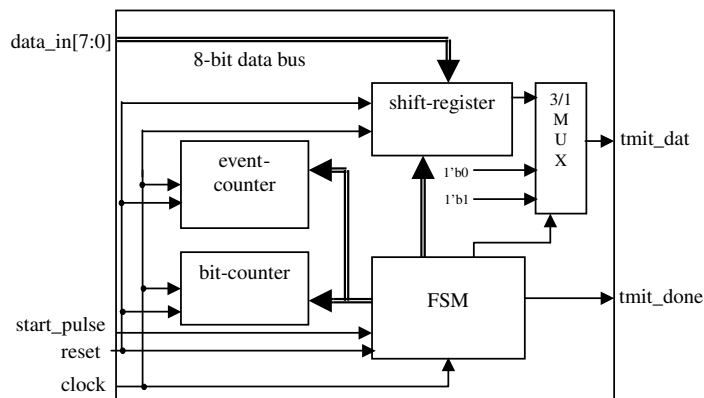


Fig. 4. Functional block diagram of the transmitter.

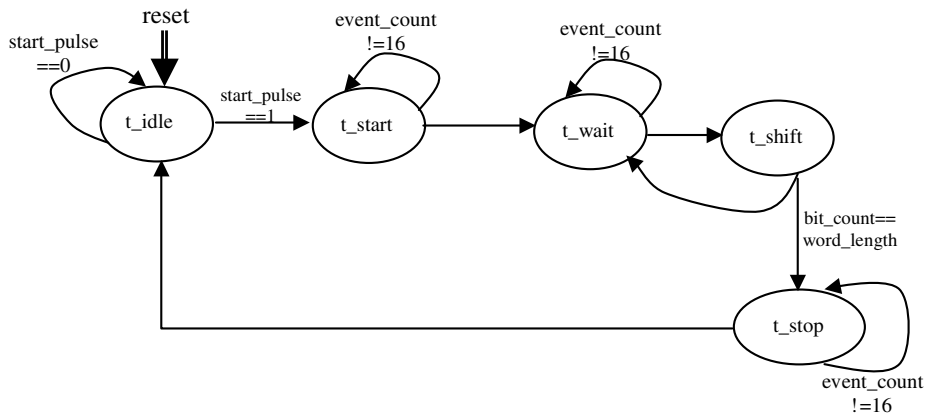


Fig. 5. State diagram of the transmitter module.

this state for 16-clock cycles and generates an active high pulse at t_done output and then jumps back to t_idle state.

Fig. 6 shows the simulation result of the transmitter module. Two time frames of simulation result (A and B) show starts and end of data transmission.

Fig. 6 shows that transmitter module starts data transmission with the active high pulse at the 'start_pulse' input while the 'reset' input is high. It can be seen that the transmission starts with logic low start-bit at 'tmit_dat' output, followed by eight data bits and one logic high stop bit. It is also seen that input data applied at the 'data_in[7:0]' (it is H'55 in the figure) has been changed into equivalent serial data stream at the serial output 'tmit_dat' (it is '01010101' in the figure) which verifies the correctness of the operation of the transmitter module. At the end of trans-

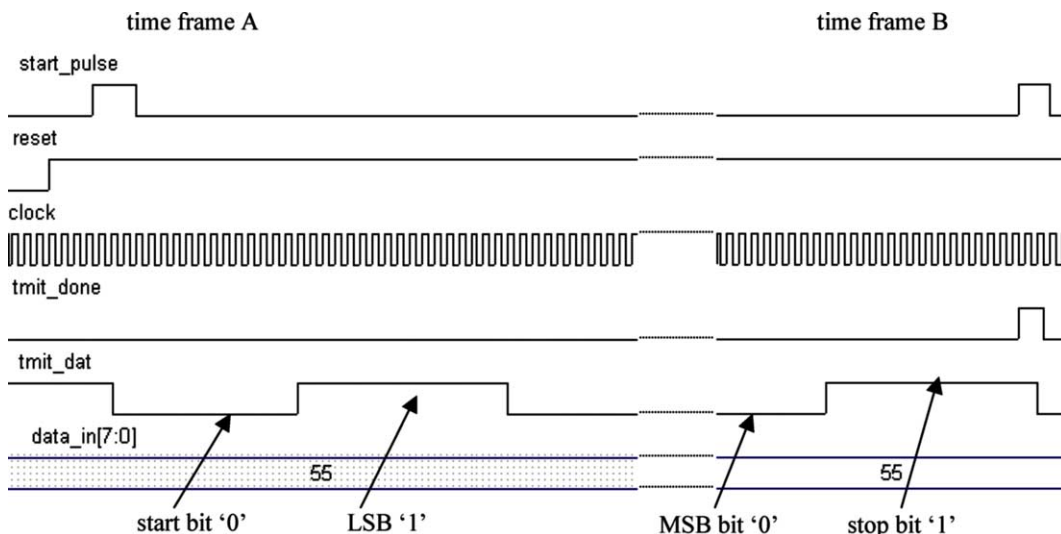


Fig. 6. Simulation result of the transmitter module of the UART.

mission of every data packet, the transmitter generates an active high pulse at the output ‘t_done’ to indicate that transmission of data packet has been performed successfully.

4.3. Receiver module

Fig. 7 shows functional block diagram of the receiver module.

It consists of the following modules:

- FSM,
- shift-register,
- bit counter,
- event counter.

FSM controls all other modules. Shift-register converts serial input data into 8-bit parallel output data. Bit counter counts number of bits transmitted and event counter counts number of clock cycles. Clock input is equal to $16 \times \text{baud_rate}$ which is generated by baud_rate generator module.

Fig. 8 shows the state diagram of the receiver module which describes its operation.

All the modules of the receiver reset upon active low signal in the ‘reset input’ and the FSM defaults to **r_idle** state where the machine idles as long as serial data input signal (s_din) goes from logic ‘1’ to ‘0’ (start-bit). Once the start-bit is detected, the FSM jumps to state **r_mid** where it goes to the centre of the start-bit by counting 8 clock cycle using event counter. Once the bit-centre is found, if s_din signal is still low then the FSM jumps to **r_wait** state otherwise it is not a valid start-bit and the FSM jumps back to **r_idle** state. Noise in the UART data line can produce this kind of effect. In the **r_wait** state, the FSM waits for 16-clock cycles and then jumps to **r_sam** state where the s_din signal is sampled into the shift-register. FSM jumps back to **r_wait** state and again waits for 16-clock cycles and then compare number of sampled-bits with that of **WORD_LEN** (presettled word-length value). If they are equal, the FSM jumps to **r_stop** state otherwise it jumps to **r_sam** state. In the **r_stop** state, the s_din signal is sensed to check for logic ‘1’ and generates

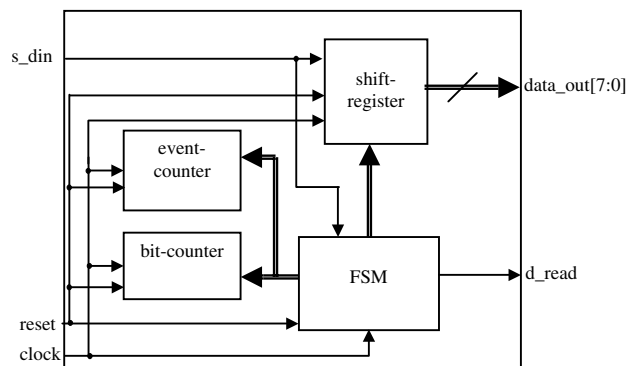


Fig. 7. Functional block diagram of the receiver module.

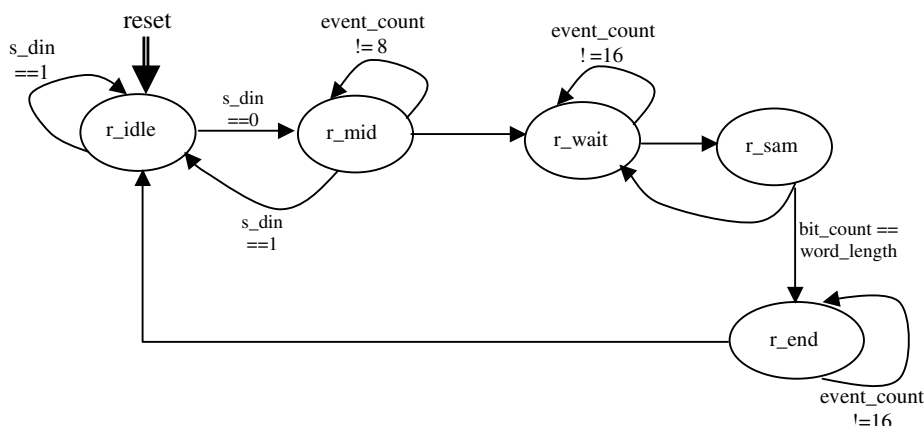


Fig. 8. State diagram of the receiver module of the UART.

active high pulse at 'd_ready' output line after waiting 16-clock cycles and then it jumps to r_idle state.

Fig. 9 shows the simulation result of the receiver module. Two time frames of simulation result (A and B) show starts and end of data receiving.

Fig. 9 shows that when reset input signal is high and the receiver gets an active low pulse in its 's_din' input, it starts data receiving. At the end of data receiving, it generates an active high pulse at the 'd_ready' output line to indicate that data is ready. Since input signal at the 's_din' is high, logic level of signal appeared at 'data_out[7:0]' shows the value H'FF, which verifies the correctness of the operation of the receiver module.

Fig. 10 shows the generated symbol of the UART where all the modules have been integrated. The design has been compiled and synthesized using MAX-PLUS II compiler. The synthesis report shows that the design of the UART needs only 48 flip-flops.

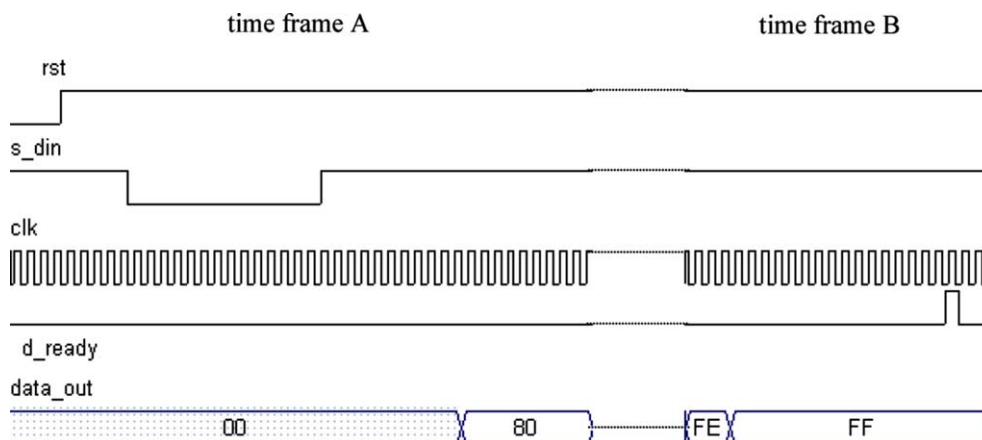


Fig. 9. Simulation result of the receiver module of the UART.

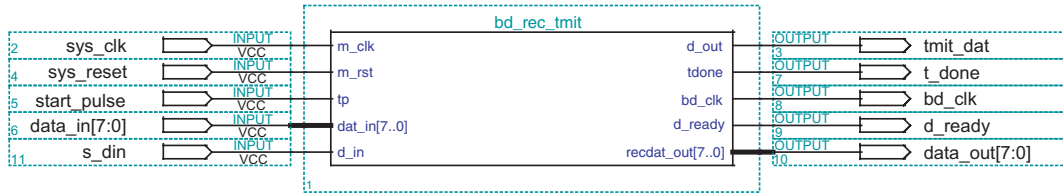


Fig. 10. Symbol of the proposed UART generated by MAX-PLUS II.

4.4. Signal description

sys_clk	This is the main system clock input. It is used to generate baud_rate
sys_reset	Main system reset input
start_pulse	Active high signal at this input starts the transmit process
data_in[7:0]	Input data that is to be sent to remote
tmit_dat	This is serialized output data for transmission to the remote
t_done	Active high signal at this input indicate end of transmission
bd_clk	Baud_clock. This is $16 \times \text{baud_rate}$
s_in	This is serial input data received from the remote
data_out[7:0]	This is the parallel output data received from the remote
d_ready	Active high signal at this input indicate data is available to read

4.5. Design verification

Fig. 11 shows simulation result for the UART where the input 's_din' has been connected with the output 'tmit_dat'. In this case, logic value of the data in the input 'data_in[7:0]' should be equal to that appeared at the output 'data_out[7:0]' at the end of data receiving.

Fig. 11 shows that the UART generates active high pulse at the output line 'd_ready' and 't_done' at the end of data transmission and receiving. The signal value (H'55) at the input data_in[7:0] and data_out[7:0] are same. It verifies the correctness of the operation of the UART.

For verification of the design at hardware, it has been downloaded to EPF10K20 (ALTERA FLEX PLD) and the signals have been checked using oscilloscope. Figs. 12 and 13 show the hardware verification results of transmitter module of the UART.

Figs. 12 and 13 show that the serial output data bits are equivalent to that of the parallel input data bits. The Figures also show a pulse that at the end of data transmission indicating that transmission has been complete. It verifies that the design of the transmitter module implemented in the hardware is functioning properly.

Logic signals as shown in the simulation results of the different modules of the UART prove its correct functionality as specified in the design requirements. It has been again verified by implementing the design in the FPGA chip. The core of the UART is configurable. The receiver and the transmitter operate independently, and each can be selectively disabled for synthesis. Synthesis reports shows that the design needs only 48 flip-flops. It proves that the size of design is extremely compact. The operation of the proposed UART follows the standard protocols of the

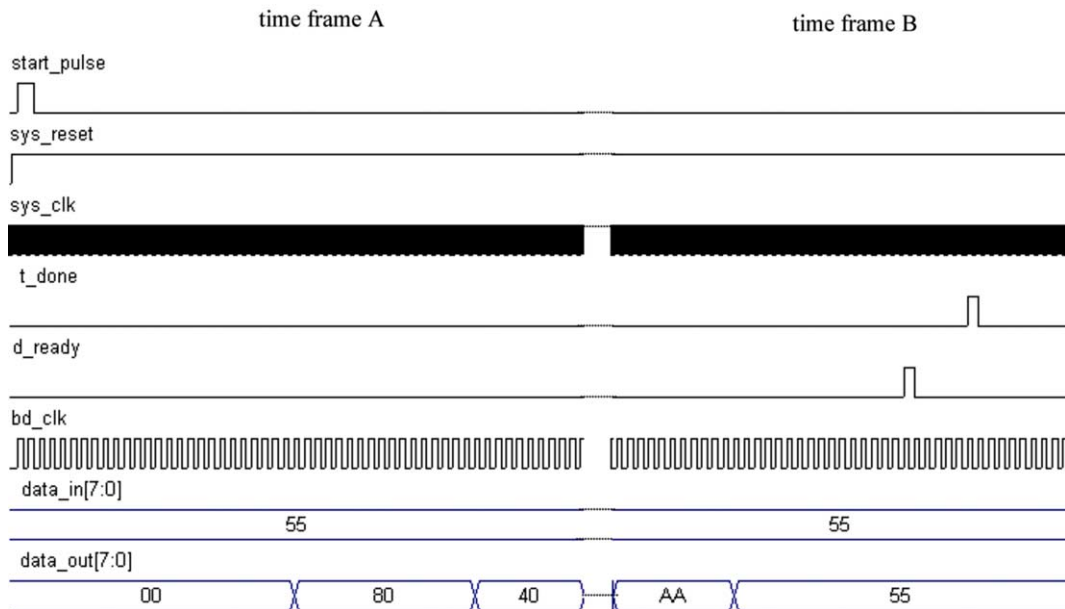


Fig. 11. Simulation result of the UART in test mode.

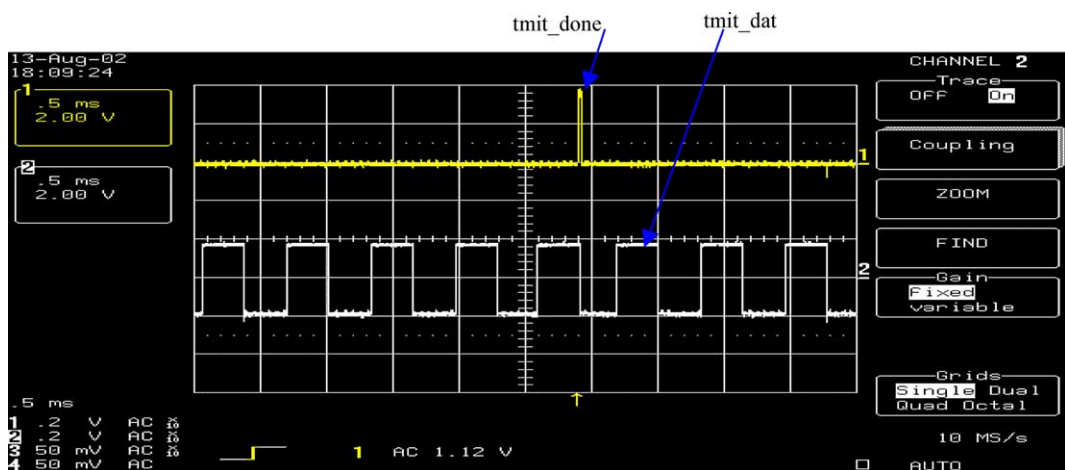


Fig. 12. Hardware verification result showing serial output (01010101) for output H'55.

serial data communication and it is compatible with any other standard UARTs such as 16550. It can be connected with any standard micro-controller and peripherals through its parallel data bus and serial data bus and can establish data communication between them. All these nice features of the proposed UART make it attractive for SoC application.

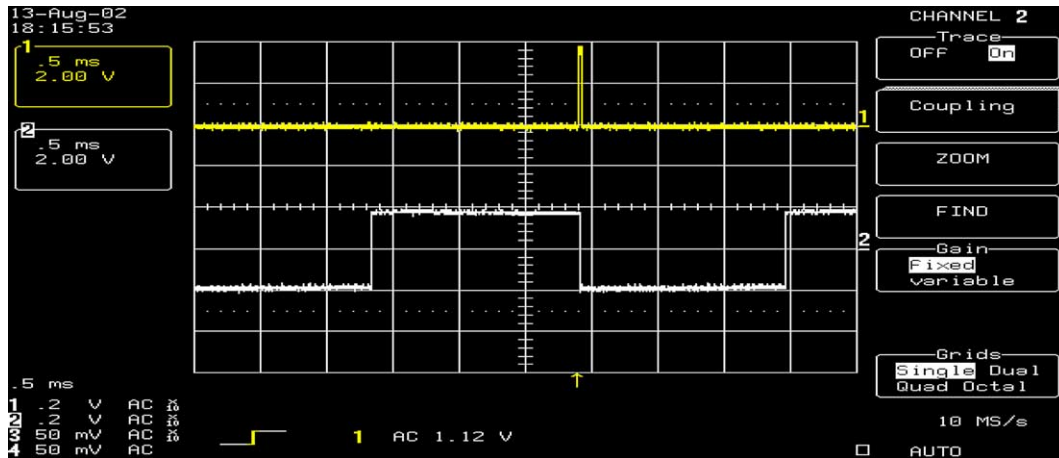


Fig. 13. Hardware verification result showing serial output (00001111) for input H'0F.

5. Conclusion

Design of a micro-UART using Verilog HDL is presented in this paper. The waveform of different signals as shown in simulation results and in oscilloscope guarantee the proper operation of the UART. Due to its compact size, programmability and configurability features, the proposed UART is ideal for SoC application. It can be used as an intellectual property. The design can be enhanced by simple modification depending on the application.

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